Rajiv Gandhi Institute of Petroleum Technology, Jais Amethi

Object-Oriented Methodologies(CS224) Quiz 2026-27

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Q.1 int a[] = {10, 20, 30, 40, 50};
int *p = a;
cout << *p++ << " ";
cout << *++p << " ";
cout << (*p)++;
What is the output?
A. 10 30 30
B. 10 20 30
C. 10 30 30
D. 10 30 31

Q.2 int a[] = {5, 10, 15, 20};
int *p = a;
cout << *p++ << " ";
cout << *p << " ";
cout << ++*p << " ";
cout << *p++;
What is the output?
A. 5 10 11 11
B. 5 10 11 15
C. 5 11 11 15
D. 5 10 10 15

Q.3 int x = 10;
int *p = &x;
int *q = p;
(*p)++;
++(*q);
cout << x << " " << *p << " " << *q;
What is printed?
A. 10 11 11
B. 12 12 12
C. 11 12 12
D. 12 11 12

Q.4 int a[] = {1, 2, 3, 4, 5};
int *p = a + 1;
cout << *p++ << " ";
cout << ++*p << " ";
cout << *--p << " ";
cout << (*p)--;
What is the output?
A. 2 4 2 2
B. 2 4 2 2
C. 2 3 2 2
D. 2 4 3 3

Q.5 void fun (int &x, int &y) {
x--;
++y;
x = x + y; }
int a = 5, b = 10;
fun(a, b); cout << a << " " << b;
What is the output?
A. 16 11
B. 15 11
C. 16 10
D. 15 10

Q.6 void fun (int &a, int &b)
{
a++;
b++; }
int x = 10;
fun (x, x);
cout << x;
What is the output?
A. 10
B. 11
C. 12
D. Undefined behavior

Q.7 struct S
{
int x; };
S a[3] = {{10}, {20}, {30}};
S *p = a;
cout << (p++)->x << " ";
cout << (p)->x << " ";
cout << (++p)->x;
Output?
A. 10 20 30
B. 10 20 20
C. 10 30 30
D. 10 30 20

Q.8 struct A
{
int x; };
A a = {10};
A *p = &a;
cout << p->x++ << " ";
cout << ++p->x << " ";
cout << p->x;
Output?
A. 10 11 11
B. 10 12 12
C. 11 12 12
D. 10 11 12

Q.9 struct Student
{
int marks; };
Student s[4]={10, 20, 30, 40};
Student *p = s + 1;
cout << (p++)->marks << " ";
cout << (++p)->marks << " ";
cout << (p--)->marks;
What is the output?
A. 20 40 40
B. 20 30 40
C. 20 40 30
D. 30 40 30

Q. 10 Which statement about an array name a and a pointer p is correct?
A. a++ and p++ are both valid if a is an integer array
B a always behaves exactly like a pointer variable
C. In most expressions, a decays to a pointer to its first element, but the array itself is not a pointer variable
D. sizeof(a) and sizeof(&a) always produce the same result.

Q.11 *p++ vs (*p)++ Which statement is correct?
A. *p++ increments the value pointed to by p.
B. (*p)++ increments the pointer p
C. *p++ is equivalent to *(p++), whereas (*p)-- increments the pointed-to value.
D. Both expressions are equivalent.

Q.12 Which of the following is NOT a valid operation on a C++ reference after it has been initialized?
A. Modifying the referenced object

Q.13 Which statement correctly describes the difference between ++x and x++?
A. Both always have identical behavior

B. Reading the referenced object C. Making the reference refer to another object D. Passing the reference to another function	B. ++x increments x after its value is used, while x++ increments before its value is used. C. ++x increments before its value is used, while x++ increments after its value is used. D. ++x can only be used with pointers.
Q.14 int a[5]; int *p = a; Which statement is correct? A. p++ increments the address by exactly 1 byte. B. p++ increments p by sizeof(int) bytes in terms of address distance. C. (*p)++ moves the pointer to the next array element. D. ++p and ++(*p) always have the same effect.	**Q.15** struct S {char c; int x;}. Why can sizeof(S) be greater than sizeof(char) + sizeof(int)? A. Because structures always contain hidden pointers B. Because of memory alignment and padding. C. Because char occupies 4 bytes inside structures. D. Because every structure contains a virtual function table

Q.16 A college maintains the academic records of N students, where each student has a roll number, name, and marks in five subjects. Develop a C++ program using an array of structures to determine the class topper, second-highest scorer, average marks, and students who are at risk of failing. A student is considered failed if marks in any subject are below 40. Implement all searching and processing operations using pointers and pointer arithmetic, use call by reference wherever student records need to be modified, and use an enum to represent the result status (Pass, Fail, Distinction). The program should also provide an option to increase the marks of all failed students in a particular subject by 5 marks (maximum 100) and then recalculate their results.